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KI-Servicezentrum Berlin-Brandenburg

RAG Forum

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Create Knowledge.**

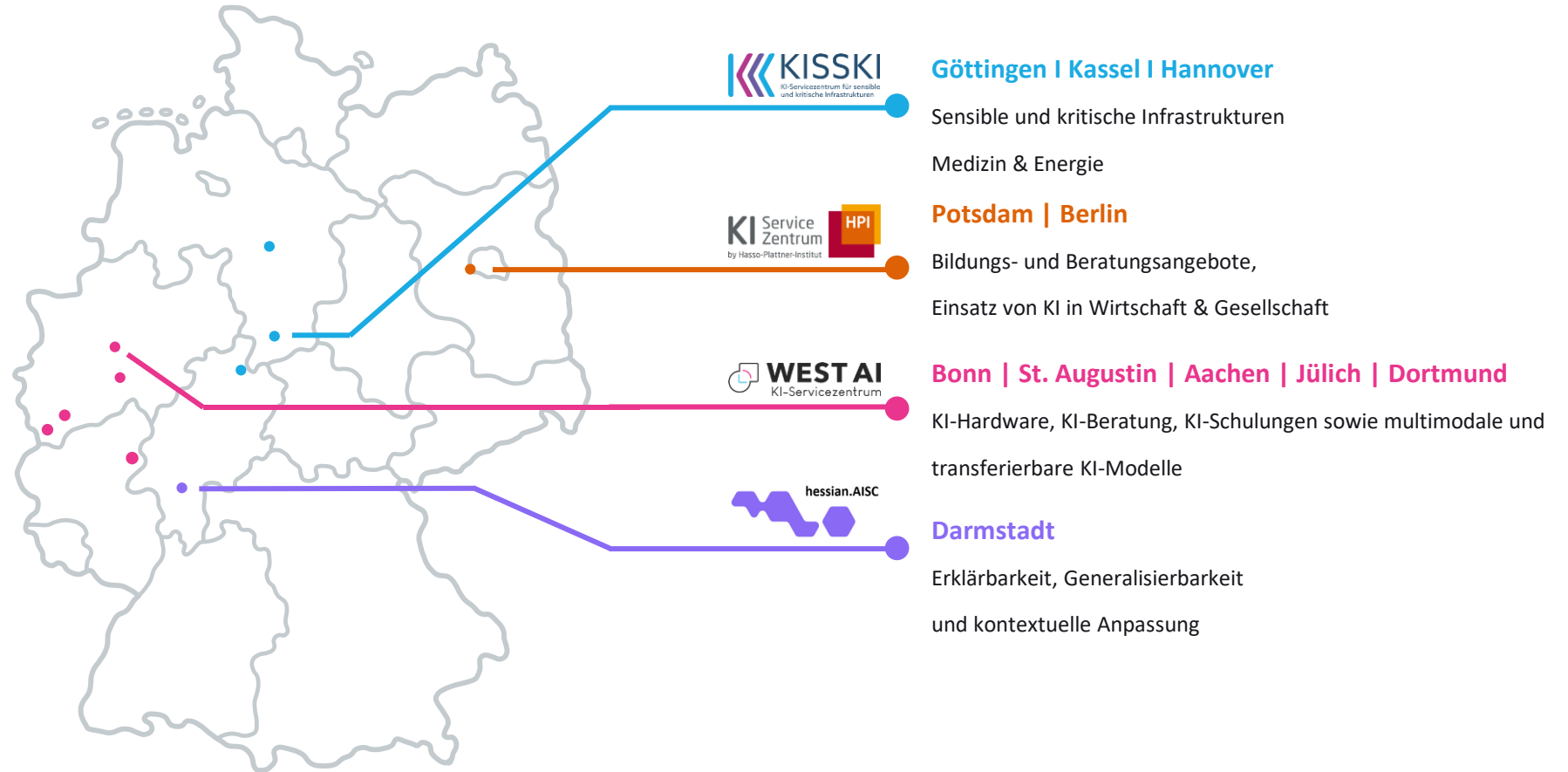
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KI-Servicezentren

Ziel:
Barrieren der
Implementierung von KI-
Anwendungen in Gesellschaft
und Wirtschaft reduzieren

ÜBERSICHT



Gefördert durch:

- KI-Methodenforschung
- KI-Betriebsforschung

PubMedCLIP: How Much Does CLIP Benefit Visual Question Answering in the Medical Domain?

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Abstract

Contrastive Language-Image Pre-training (CLIP) has shown remarkable success in learning with cross-modal supervision from extensive amounts of image-text pairs collected online. Thus far, the effectiveness of CLIP has been investigated primarily in general-domain multimodal problems. In this work, we evaluate the effectiveness of CLIP for the task of Medical Visual Question Answering (MedVQA). We present PubMedCLIP, a fine-tuned version of CLIP for the medical domain based on PubMed articles. Our experiments conducted on two MedVQA benchmark datasets illustrate that PubMedCLIP achieves superior results improving the overall accuracy up to 3% in comparison to the state-of-the-art Model-Agnostic Meta-Learning (MAML) networks pre-trained only on visual data. The PubMedCLIP model with different back-ends, the source code for pre-training them and reproducing our MedVQA pipeline is publicly available at <https://github.com/sarahESL/PubMedCLIP>.

1 Introduction

Medical visual question answering (MedVQA) seeks answers to natural language questions about a given medical image. The development of MedVQA has considerable potential to benefit health care systems, as it may aid clinicians in interpreting medical images and obtaining more accurate diagnoses by consulting a second opinion. Thus, it has become a very active area of research, with competitive benchmarks and yearly competitions (Abacha et al., 2021). Yet, visual question answering in the medical domain in particular remains non-trivial as we suffer from a general lack of large balanced training data, in part due to privacy concerns. To solve the multimodal task of MedVQA, a system must understand both medical images and textual questions and infer the associations between them sufficiently well to produce a correct answer (An

Findings of the Association for Computational Linguistics
May 2-6, 2023 ©2023 Association for Computational Linguistics

Exploring Paracrawl for Document-level Neural Machine Translation

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Abstract

Document-level neural machine translation (NMT) has outperformed sentence-level NMT on a number of datasets. However, document-level NMT is still not widely adopted in real-world translation systems mainly due to the lack of large-scale general-domain training data for document-level NMT. We examine the effectiveness of using Paracrawl for learning document-level translation. Paracrawl is a large-scale parallel corpus crawled from the Internet and contains data from various domains. The official Paracrawl corpus was released as parallel sentences (extracted from parallel webpages) and therefore previous works only used Paracrawl for learning sentence-level translation. In this work, we extract parallel paragraphs from Paracrawl parallel webpages using automatic sentence alignments and we use the extracted parallel paragraphs as parallel documents for training document-level translation models. We show that document-level NMT models trained with only parallel paragraphs from Paracrawl can be used to translate real documents from TED, News and Europarl, outperforming sentence-level NMT models. We also perform a targeted pronoun evaluation and show that document-level models trained with Paracrawl data can help context-aware pronoun translation. We release our data and code here¹.

1 Introduction

The Transformer translation model (Vaswani et al., 2017), which performs sentence-level translation based on attention networks, has achieved great success and significantly improved the state-of-the-art in machine translation. Compared to sentence-level translation, document-level translation (Xu et al., 2021; Bao et al., 2021; Jauregi Unanue et al., 2020; Ma et al., 2020; Maruf et al., 2019; Tu et al., 2018; Maruf and Haffari, 2018) performs translation at document-level and can potentially fur

¹<https://github.com/Yusser96/Exploring-Paracrawl-for-Documents-level-Neural-Machine-Translation>

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Efficient Parallelization Layouts for Large-Scale Distributed Model Training

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Abstract

Efficiently training large language models requires parallelizing across hundreds of hardware accelerators and invoking various compute and memory optimizations. When combined, many of these strategies have complex interactions regarding the final training efficiency. Prior work tackling this problem did not have access to the latest set of optimizations, such as FLASHATTENTION or sequence parallelism. In this work, we conduct a comprehensive ablation study of possible training configurations for large language models. We distill this large study into several key recommendations for the most efficient training. For instance, we find that using a micro-batch size of 1 usually enables the most efficient training layouts. Larger micro-batch sizes necessitate activation checkpointing or higher degrees of model parallelism and also lead to larger pipeline bubbles. Our most efficient configurations enable us to achieve state-of-the-art training efficiency results over a range of model sizes, most notably a Model FLOPs utilization of 70.5% when training a LLAMA 13B model.

1 Introduction

The number of parameters and computational resources spent on training deep neural networks is growing rapidly [1, 3, 14]. The largest models consisting of hundreds of billions of parameters do not even fit onto a single hardware accelerator. Thus, training these models requires various ways of reducing the memory requirements, such as ZeRO [16], activation checkpointing [2], and 3D-parallel (data, tensor, and pipeline parallel) training [13]. 3D parallelism, in particular, has been demonstrated to be effective for the training of Transformer-based large language models (LLMs) with hundreds of billions of parameters [13].

However, training these models efficiently with 3D parallelism requires significant domain expertise and extensive manual effort to determine the ideal configurations. These configurations not only need to combine data, model, and pipeline parallelism most efficiently, but also consider complex interactions with other memory and compute optimizations. FLASHATTENTION [5] in particular has had a notable impact since its release, enabling us to train models at previously impossible degrees of training efficiency. In light of these developments, we conduct a systematic study via a large-scale training efficiency sweep of these interactions. We consider up to 256 GPUs and LLAMA models with up to 65 billion parameters.

Workshop on Advancing Neural Network Training at 37th Conference on Neural Information Processing Systems (WANT@NeurIPS 2023).

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BILDUNG



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tele-task.de/series/1463

- Gastvorträge zu Forschung und Innovation

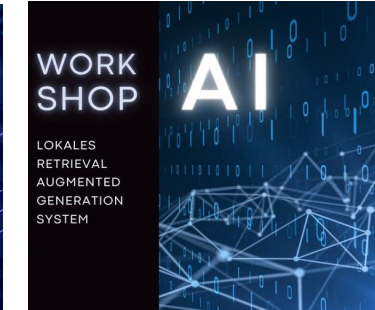


Workshops



aimaker.community

- Praxisnahe Themen
- Beispielthemen: Speech2summary, Docker für ML, semantische Suche



MOOCs



open.hpi.de/channels/ai-service-center

- ChatGPT: Was bedeutet generative KI für unsere Gesellschaft?
- Profitable KI
- KI Biases verstehen und vermeiden



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 - Daten sollten **anonymisiert** oder **synthetisiert** sein
 - Kein **Hosting** von Produkten
- **Reporting** & **Veröffentlichung** durch Nutzende
- **Altrechte** bleiben bei Nutzenden
- **Neurechte** bleiben bei Nutzenden
 - Einräumen von Nutzungsrechten für Forschung und Lehre

Training

- 64 NVIDIA H100 GPU

Inferenz

- 40 NVIDIA A30 GPU

ARM Server

- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server

- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge

- ARMv8 CPU
- NVIDIA Jetson AGX Module

Neuromorph

- 288 SpiNNaker2 Chips

Speicher

- 1.5 PB NVRAM

Netzwerk

- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

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BERATUNG



Sprechstunde buchen

KI-Sprechstunde

- KI-Infrastruktur
- KI-Modelle & Frameworks
- KI-Anwendungsfälle

KI-Pilotprojekte

- Co-Entwicklung eines Prototyps
- Bewerbung alle drei Monate
- Auswahlkriterien z.B. KI-Reife, Gemeinwohl
- Veröffentlichung der Ergebnisse

Kooperationen

- Gemeinsam organisierte Netzwerktreffen



Bisherige KI-Pilotprojekte

- Generierung Mathematik-Problemen
- Generierung Leichter Sprache
- Generierung von Upcycling Vorschlägen
- Reduzierung von Food Waste
- Datierung mittels Handschrift



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PILOTPROJEKT: GENERATIVE KI



huggingface.co/datasets/aihpi/fashion-prompt-to-prompt



A cropped jacket with patch pockets, featuring bold tribal prints and tassel closures.

Original Caption



Add a detachable faux fur collar to the neckline for winter wear.

Edit Instruction

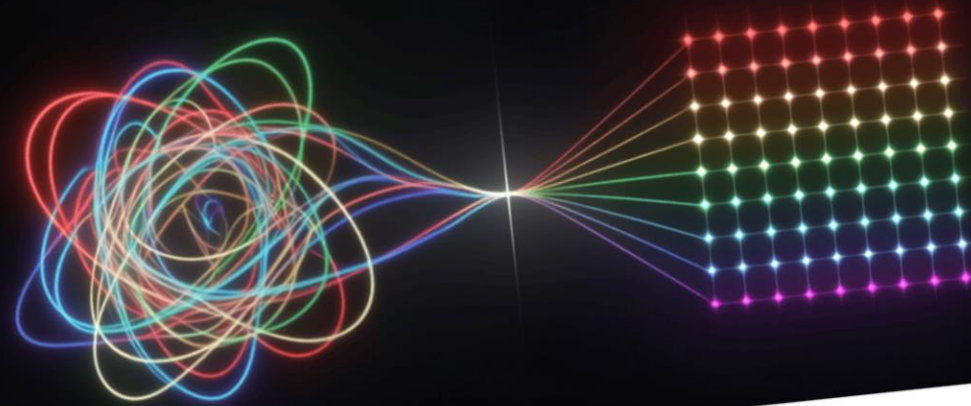


A cropped jacket with patch pockets, featuring bold tribal prints and tassel closures, now includes a detachable faux fur collar at the neckline.

Resulting Caption

turbovec

Google's TurboQuant
for vector search



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EMBEDDINGS ARE BIG. BIG IS SLOW

- **Example:**
Embed **100k-chunks** of German Wikipedia with **minilm-embedding**
- Each text chunk is stored as one float32 vector: **384 numbers × 4 bytes = 1.5 KB**
- Our 100k-chunk slice of German Wikipedia already needs **154 MB**;
→ **100M chunks would be ~150 GB**
- These vectors must sit in **RAM** for fast search
→ memory is both the bottleneck and the bill
- Exact float32 search already costs **3.7 ms/query (~271 q/s)** on just 100k vectors.

→ **Can we shrink the vectors without wrecking retrieval quality?**

JUST A LIST OF NUMBERS

- Embeddings are coordinates that place **similar meanings** close together.
- Search means:
find the chunks whose coordinates sit **nearest** to the question's coordinates.
- Float32 embeddings have about **7 digits of precision**
- That is like writing every measurement to 7 decimals when you **only need one**
→ most of those bits **never change the answer**

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QUANTISATION

- **turbovec** keeps only a handful of levels per number:
 - 2 bits = 4 levels
 - 3 bits = 8
 - 4 bits = 16
- A per-vector scale places those few levels where that vector's values actually lie, e.g.
 - original vector: $[-0.7, -0.18, 0.51, 0.8]$
 - range: $0.8 - (-0.7) = 1.5$
 - 2 bit spacing: $1.5 / (4-1) = 0.5$ (assuming uniform distribution)
 - quantised vector: $[-0.7, -0.2, 0.3, 0.8]$
- **turbovec** uses precomputed quantisation levels that are not uniformly distributed)

32 bits $\rightarrow$ 2 bits is **16× fewer bits** per number

ROUNDING DOESN'T BREAK SEARCH

- To rank nearest neighbours you only need **relative positions**, not exact values
- Rounding nudges every point a little but keeps it in roughly the **same place**
→ **neighbours stay neighbours**
- So the right answers still land near the top:
at 4-bit we keep **93%** of them, at 2-bit **82%**
- Want the last bit of quality back?
→ **Re-rank just the top hits** using the original float32 vectors.

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SMALLER IS ALSO FASTER

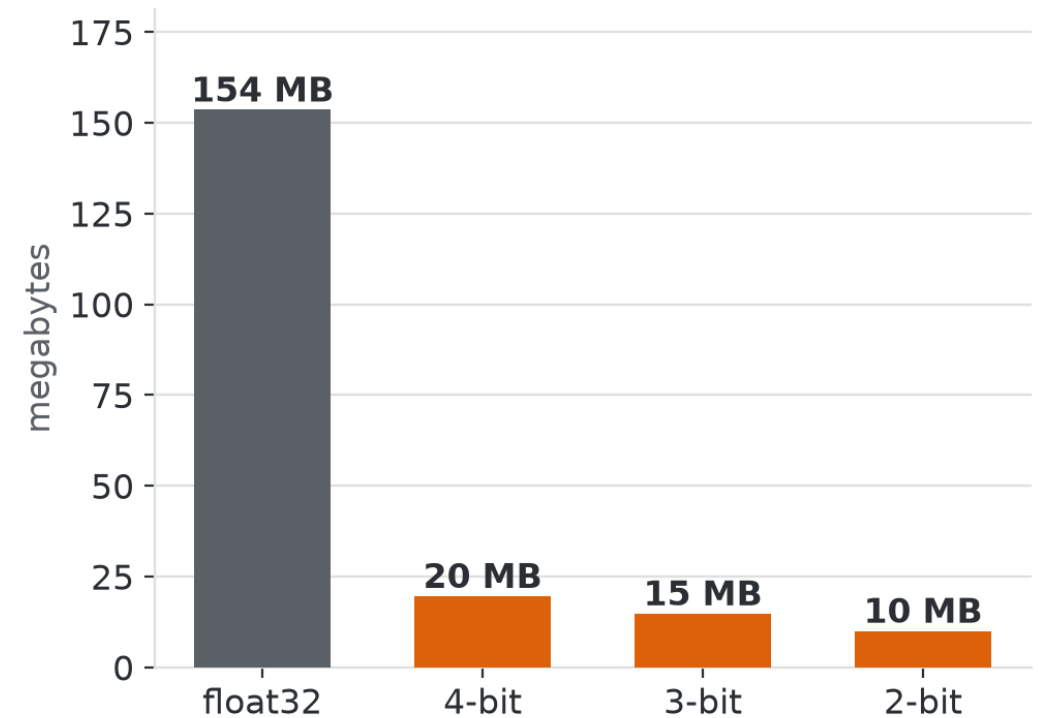
- Searching means **reading** every stored vector and **comparing** it to the query
→ work that is limited mostly by **memory speed**
- Smaller codes mean far less data to move, so search **speeds up with the shrink**
- *turbovec* compares the **packed codes** directly
→ it never unpacks them back to float32 first
- It leans on special CPU instructions that crunch **many numbers at once**
→ **Single Instruction, Multiple Data (SIMD)**: AVX-512 on x86, NEON on ARM)
(different from GPU tasks: Multiple Instruction, Single Data)

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UP TO 15× SMALLER

- Data: 100k chunks, dim 384
- **float32: 154 MB** (Baseline)
- **4-bit: 19.6 MB (7.8× smaller)**,
recall@10 = **0.934**
- **3-bit: 14.8 MB (10.4× smaller)**,
recall@10 = **0.897**
- **2-bit: 10.0 MB (15.4× smaller)**,
recall@10 = **0.816**

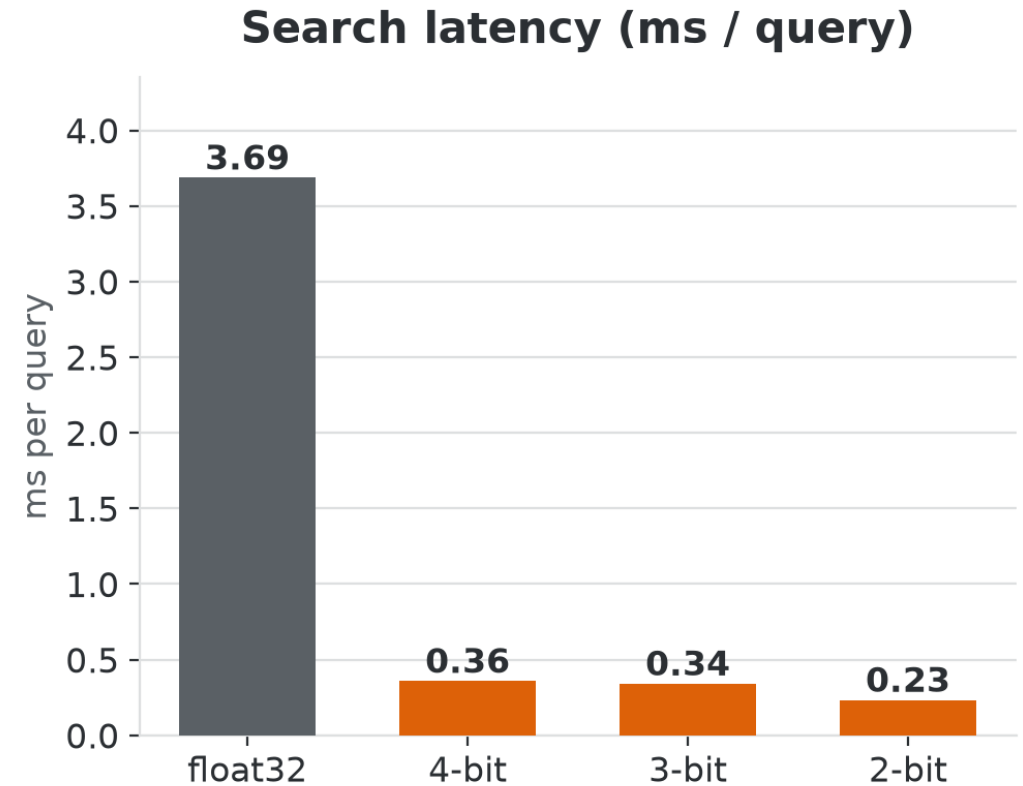
Index size on disk



Gefördert durch:

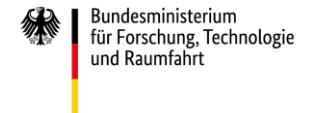
UP TO 16× FASTER

- **Exact float32 search:**
3.69 ms/query (**~271 q/s**)
- **4-bit:**
0.36 ms/query (10.3× faster, **~2791 q/s**)
- **3-bit:**
0.34 ms/query (10.8× faster, **~2931 q/s**)
- **2-bit:**
0.23 ms/query (15.8× faster, **~4299 q/s**)



Gefördert durch:

Gefördert durch:



Turbovec:
<https://arxiv.org/pdf/2504.19874>

Repository for RAG forum:
<https://github.com/aihpi/workshop-rag-forum>

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hpi.de/kisz

**Ihre
Meinung ist
uns wichtig!**



QR-Code zum Feedback-
Formular



THREE QUESTIONS

- Embedding search is often treated as a solved problem. In your experience, what is the bigger constraint in production RAG systems: retrieval quality, retrieval speed, infrastructure costs, or something else? Where would you invest engineering effort?

Gefördert durch:

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- Which component of your RAG stack has delivered the greatest improvement relative to the effort required? Why was it such a "game changer"?

Gefördert durch:

THREE QUESTIONS

- Embedding search is often treated as a solved problem. In your experience, what is the bigger constraint in production RAG systems: retrieval quality, retrieval speed, infrastructure costs, or something else? Where would you invest engineering effort?
- Which component of your RAG stack has delivered the greatest improvement relative to the effort required? Why was it such a "game changer"?
- How do you currently determine whether your RAG system is "good enough"? What metrics, benchmarks, or user feedback do you trust most? What types of failures in RAG systems are most difficult to detect before users encounter them?

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